

Solution Requirements

Date	29 june 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system must allow bank administrators and credit officers to securely log into the risk dashboard using standard username and password credentials to prevent unauthorized data access.</i>	Medium
2	Authorization levels	<i>The system must support two user roles: Admin (can upload new raw CSV training sets and retrain models) and Credit Officer (can only use the form layout to input</i>	High

		<i>applicant data and fetch prediction results).</i>	
3	External interfaces	<i>The system must be capable of reading local file data inputs (application_record.csv and credit_record.csv) and outputting a serialized binary pipeline structure (model.pkl).</i>	High
4	Transactions processing	<i>The backend Flask server must capture client demographic data payload dictionaries from form submissions, convert categorical text fields via LabelEncoder, and process the arrays synchronously through the classifier.</i>	High
5	Reporting	<i>The application must render a summary results view (result.html) printing a definitive status message indicating whether the specific application is "Approved" or "Rejected" alongside back-to-home navigation links.</i>	High
6	Business rules, etc.	<i>If an applicant has a delinquency tracking status of 1 to 5 (more than 30 days overdue) during any historical monitoring month, the system's group aggregation logic must automatically flag their risk status profile as 0 (Rejected).</i>	High

7	Compliance to laws or regulations	<i>The preprocessing layer must exclude sensitive personal identity flags (such as dropping specific explicit keys or keeping high-level demographic indicators anonymized) to comply with local financial privacy guidelines and fair lending practice acts.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	<i>The system must process the uploaded profile data and return the final approval or rejection verdict on the browser screen almost instantaneously.</i>	<i>The Flask web engine must render the result.html template in under 1.5 seconds from the moment the user clicks the "Predict" button.</i>
2	Scalability	<i>The backend prediction route logic must handle parallel operations effectively without locking the binary scoring model file asset</i>	<i>The application must support up to 50 concurrent model queries simultaneously without memory leaks or server crashes.</i>
3	Security & Data Privacy	<i>The application folder and serialized model pipeline file (model.pkl) must be secured within local environments, preventing unauthorized code execution or code tampering.</i>	<i>Passwords for the system must be hashed using secure libraries (e.g., bcrypt), and the input forms must validate strings to avoid SQL or script injections.</i>

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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4	Reliability & Availability	<i>The predictive local server stack must remain up and functional for credit processing teams during business hours.</i>	<i>The application must maintain a standard target uptime metric of 99.9% availability during active operational shifts.</i>
5	Usability & Accessibility	<i>The user interface form must be completely clean, intuitive, and easy to operate without intensive specialized technical training.</i>	<i>Built using clean Bootstrap HTML components, ensuring that new credit operators can completely fill the fields and generate accurate predictions within a single 2-minute session.</i>